

POCKETSMART AI: TECHNICAL COMMUNICATION & PRESENTATION REPORT

1. Executive Summary & Communication Objectives

Technical communication in software engineering bridges complex backend implementations with clear stakeholder comprehension. For **PocketSmart AI**, the communication strategy was engineered to articulate the technical synergy between asynchronous FastAPI controllers, Google Gemini generative AI prompts, and real-time e-commerce affiliate routing to academic evaluators, peers, and non-technical end-users.

2. Stakeholder Engagement Matrix

Stakeholder Group	Primary Objective	Communication Channel	Technical Depth Level	Key Focus Areas
Academic Evaluators & Examiners	Project assessment, architectural defense, code verification.	Structured viva defense, code walkthrough, milestone PDFs.	Deep Architectural & Code-Level	FastAPI async design, JWT auth, fail-safe algorithmic fallback, API latency.
Development Team Members	Task distribution, code integration, debugging	Daily standups, Git commit notes, collaborative review sessions.	Implementation & Code-Level	Routing decoupling, schema models, JSON

	synchronizati on.			persisten ce integrity.
End Users & Project Audience	Usability demonstratio n, practical utility validation.	Video walkthrough (demo vid link.txt), interactive UI testing.	Functional & Solution- Centric	Budget inputs, ease of shopping redirectio n, instant plan accuracy.

3. Communication Channels & Collaborative Workflows

- **Sprint Standups & Milestone Checkpoints:** Regular synchronization sessions ensured continuous integration of frontend Jinja2 templates with backend routers without circular dependency bottlenecks.
- **Code Review & Inline Documentation:** Every critical module (routes/planners.py, app.py, auth.py) incorporates descriptive inline docstrings and standardized HTTP status definitions for clear peer readability.
- **Viva Voce Defense Strategy:** Prepared structured responses for evaluator queries covering token lifespan, password security, API quota limits, and database selection criteria.

4. Technical Articulation & Presentation Delivery Standards

To convey high technical rigor during evaluations, complex features were mapped to definitive terminology:

- *Deterministic Fallback:* Articulated as a "heuristic business-logic shield preventing 500 runtime exceptions during external LLM throttling."

- *Data Persistence*: Explained as an "atomic file-based persistent JSON datastore abstracting heavyweight relational overhead for optimized local deployment."
- *Deep-Linking*: Defined as "dynamic URL-encoded affiliate query builders targeting premier Indian retail endpoints."